NIKHIL YARRA

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LinkedIn | GitHub | Portfolio

RESEARCH INTERESTS

Reliable LLM-based agents; evidence-grounded retrieval and generation; tool-observation validation, provenance and execution assurance; reproducible evaluation of system behaviour and failure modes.

EDUCATION

M.S. Data Science — New Jersey Institute of Technology, USA — Sep 2023 – Dec 2024

Coursework: machine learning, AI, deep learning, reinforcement learning, applied statistics, big data.

B.Tech. Computer Science & Engineering — GITAM, Hyderabad, India — Jun 2019 – May 2023

Coursework: AI, data mining, databases, cloud computing, robotics and automation.

RESEARCH OUTPUTS

Tool-Observation Integrity in AI Agents: A Critical Survey of Validation, Provenance, and Execution Assurance.

SSRN | doi:10.2139/ssrn.7545819 — 30 Sep 2026

Simulator Update Frequency Sensitivity in Driving Scenario Search.

Zenodo | doi:10.5281/zenodo.23817382 — 31 Oct 2026

SELECTED RESEARCH & ENGINEERING PROJECTS

ModelGuard — LLM Regression Evaluation *Python | Pydantic | SQLite | GitHub Actions*

- Built reproducible prompt-version comparisons with paired regression analysis, category-level metrics, exact McNemar tests, Wilson intervals and rolling drift monitoring.
- Implemented CI reports and pull-request summaries; validated software behaviour with 93 automated tests, Python 3.11–3.13 and Docker. Live model and human-reviewed gold-set evaluation pending.

DocAtlas — Evidence-First Retrieval & Evaluation *Python | FastAPI | SQLite FTS5 | FastEmbed*

- Built document ingestion and persistent indexing with BM25, dense and hybrid retrieval; added source-ID and exact-quote citation validation.
- Compared retrieval methods using reproducible evaluations, category-level metrics, failure reports and regression checks.

DocMend — Evidence-Bound Documentation Maintenance *Python | Git | Docker | GitHub Actions*

- Compared immutable Git revisions and mapped Python code changes to Markdown sections; generated drift findings with supporting evidence and scoped repair proposals.
- Enforced static-evidence checks before applying safe edits; verified a controlled stale-signature repair against a DocAtlas snapshot. Built a report-import review dashboard.

EXPERIENCE

AI Engineer — **Jan 2026 – Sep 2026**

Warren and Carter Technologies Inc., USA

- Built LLM and conversational AI workflows combining retrieval, multi-step execution, tool routing and structured outputs.
- Implemented validation, fallbacks and human-review checkpoints; tested task variations and failure cases to refine retrieval, routing and execution.

Data Analyst — **Apr 2025 – Feb 2026**

Tubman Technologies Inc., USA

- Developed Python and SQL workflows for healthcare and financial data preparation, validation and analysis; processed documents for retrieval and produced Power BI/Tableau views.

Data Analyst — **Jul 2022 – Jul 2023**

SYMNN, Hyderabad, India

- Analysed telecom network and GIS data, built predictive models, and prepared validated datasets for utilisation and capacity-planning analysis.

Data Engineer Intern — **May 2022 – Jun 2022**

Phoenix Global, Hyderabad, India

- Led a five-member analytics project team; contributed to cleaning, analysis, Tableau/Power BI dashboards and presentation of findings.

TECHNICAL SKILLS

AI / Retrieval	RAG, embeddings, BM25, dense and hybrid retrieval, agent workflows, tool calling, structured outputs, evaluation, citation validation
Engineering	Python, SQL, FastAPI, REST APIs, Git, automated testing, Docker, GitHub Actions, SQLite/FTS5, Pydantic
ML / Analytics	TensorFlow/Keras, scikit-learn, Pandas, NumPy, transfer learning, ETL, data validation, Power BI, Tableau